OPTIMIZATION ENGINEERING

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Assignment - 1 CO-1

SET - A

(1.) Minimize $g(x) = x^2 + \frac{25}{x}$. Using the exhaustive search method with the search space. lower bound: $u = 0.5$; upper bound: $v = 5.5$ and number of equal divisions: $n = 1$

Given,

$$g(x) = x^2 + \frac{25}{x}, \quad [0.5, 5.5], \quad n = 10$$

$$\Delta x = \frac{b-a}{n} = \frac{5.5 - 0.5}{10} = \frac{5}{10} = \frac{1}{2} = 0.5$$

1) $x_1 = 0.5 \Rightarrow g(x_1) = g(0.5) = 50.25$

$x_2 = x_1 + \Delta x \Rightarrow 0.5 + 0.5 = 1 \Rightarrow g(x_2) = g(1) = 26$

$x_3 = x_2 + \Delta x = 1 + 0.5 = 1.5 \Rightarrow g(x_3) = g(1.5) = 18.85$

$\therefore g(x_1) > g(x_2) > g(x_3)$

2) $x_1 = 1 \Rightarrow g(x_1) = g(1) = 2.6$

$x_2 = 1.5 \Rightarrow g(x_2) = g(1.5) = 18.85$

$x_3 = x_2 + \Delta x = 1.5 + 0.5 = 2 \Rightarrow g(x_3) = g(2) = 16.5$

$\therefore g(x_1) > g(x_2) > g(x_3)$

3) $x_1 = 1.5 \Rightarrow g(x_1) = g(1.5) = 18.85$

$x_2 = 2 \Rightarrow g(x_2) = g(2) = 16.5$

$x_3 = x_2 + \Delta x = 2 + 0.5 = 2.5 \Rightarrow g(x_3) = g(2.5) = 16.25$

$\therefore g(x_1) > g(x_2) > g(x_3)$

4) $x_1 = 2 \Rightarrow g(x_1) = g(2) = 16.5$

$x_2 = 2.5 \Rightarrow g(x_2) = g(2.5) = 16.25$

$x_3 = x_2 + \Delta x = 2.5 + 0.5 = 3 \Rightarrow g(x_3) = g(3) = 17.3$

$\therefore g(x_1) > g(x_2) < g(x_3)$

min $\in (x_1, x_3)$

$(2, 3)$

$\therefore$ Pareto - Front (Train, Flight)

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(2) Naveen and his friends ...

(2) Identify and classify the critical path of 2310040070

$f(x) = x^5 - 4x^4 + 3x^3 + 2$

Given,

$$f(x) = x^5 - 4x^4 + 3x^3 + 2$$

$$\nabla f(x) = \frac{\partial f}{\partial x} = 5x^4 - 16x^3 + 9x^2$$

$$= x^2(5x^2 - 16x + 9)$$

$$x^2(5x^2 - 16x + 9) = 0$$

$$x_1 = 0$$

$$5x^2 - 16x + 9 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{16 \pm \sqrt{256 - 4 \cdot 5 \cdot 9}}{2 \cdot 5}$$

$$= \frac{16 \pm \sqrt{76}}{10}$$

$$= \frac{16 \pm 2\sqrt{19}}{10}$$

$$= \frac{8 \pm \sqrt{19}}{5}$$

$$\therefore x_1 = 0, \; x_2 = \frac{8 - \sqrt{19}}{5}, \; x_3 = \frac{8 + \sqrt{19}}{5}$$

$$\nabla^2 f = 20x^3 - 48x^2 + 18x \Big|_{x_1 = 0}$$

$$= 20(0)^3 - 48(0)^2 + 18(0)$$

$$= 0$$

$$\nabla^3 f = 60x^2 - 96x + 18 \Big|_{x=0}$$

$$= 60(0) - 96(0) + 18$$

$$= 18 \rightarrow \text{non-zero}$$

$\therefore$ Inflection point

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③ Naveen and his friends are planning a goa trip during summer. They prefer maximum comfort and speed during their journey. Based on table given below. Identify the pareto-front and give them the best possible journey options for the Goa trip.

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option	speed (score 1-10)	comfort (score -10)
A (Bus)	4	3
B (Train)	7	8
C (Flight)	9	6
D (carpool)	4	5
E (walking)	1	1

Ans) 1)

	Bus	Train
speed	4	7
comfort	3	8

Train dominates Bus is eliminated

2)

	Train	Flight
Speed	7	9
comfort	8	6

Non-dominated set take either train or flight

3)

	Flight	carpool
speed	9	4
comfort	6	5

Flight dominated carpool carpool is eliminated

4)

	Flight	walking
speed	9	1
comfort	6	1

Flight dominates walking walking is eliminated.

∴ Pareto-Front (Train, Flight)

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(4) Ravi Teja has 25 hectares of land for growing wheat and corn. The cost per hectares for wheat is Rs. 40, and for corn is Rs. 30. He has a total budget of Rs. 600. Wheat requires 2 man-days per hectare and corn requires 3 man days per hectare. There are 50 man-days available. The profit on wheat is Rs. 120 per hectare and on corn is Rs. 80 per hectare. Help Ravi to identify the number of hectares of each crop he should sow to maximize profit.

let, wheat $= x$, corn $= y$

$P = 120 + 80y$

constraints:-

$x + y \leq 25$

$40x + 30y \leq 600$

$2x + 3y \leq 50$

$x + y \leq 25$

$x + y = 25$

$x = 25$

$y = 25$

$(25, 25)$

$40x + 30y \leq 600$

$40x + 30y = 600$

x-intercept

$40x = 600$

$x = 15$

y-intercept

$30y = 600$

$y = 20$

$(15, 20)$

$2x + 3y \leq 50$

$2x + 3y = 50$

x-intercept

$2x = 50$

$x = 25$

y-intercept

$3y = 50$

$y = 50/3$

$(25, 50/3)$

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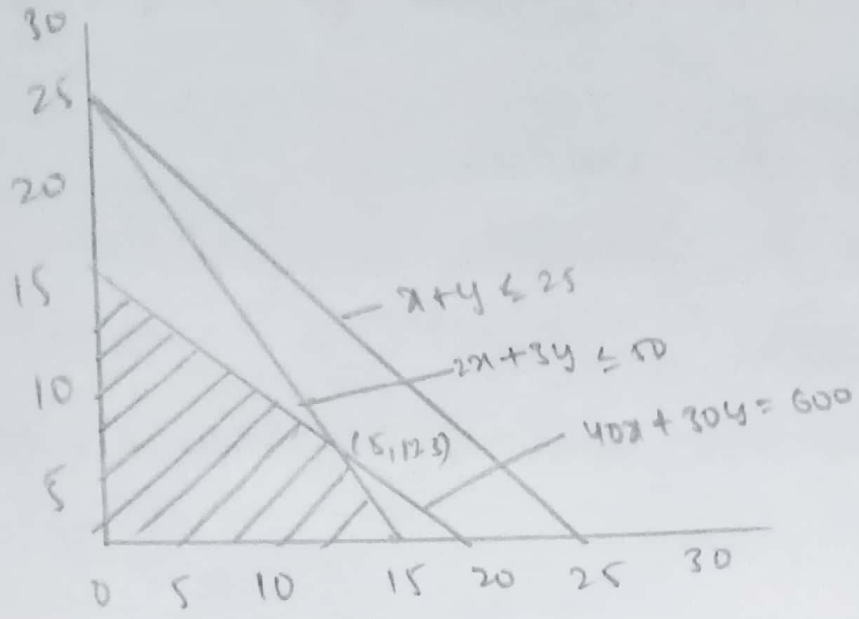

$2x+3y=50$

$2x = 50-3y$

$x = \frac{50-3y}{2}$

$x = \frac{50-3(40/5)}{2}$

$x = \frac{50-40}{2} = \frac{10}{2}$

$\boxed{x=5}$

$40x + 30y = 600$

$40\left(\frac{50-3y}{2}\right) + 30y = 600$

$1000 - 60y + 30y = 600$

$1000 - 30y = 600$

$30y = 400$

$\boxed{y = \frac{40}{3}}$

	$120x + 80y$
(0,0)	0
(15,0)	1800
(5, 13.3)	1664
(0,15)	1200

$\rightarrow$ max $120x + 30y$ at (15,0)

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(5) Describe the steps involved in optimal problem formulation.

* Clearly define the objective
  Decide what to maximize or minimize
* Identify decision variables
  Select controllable variables affecting the outcome
* Specify constraints
  Express all limits mathematically
* Simplify and linearize.
  Reduce complexity where possible
* Check for feasibility
  Ensure constraints allow a solution
* Normalize and scale
  Adjust values for numerical stability.
* Consider multiple objectives
  Balance trade-offs between goals
* Validate the formulation
  Ensure the model reflects reality
* Refine and iterate
  Improve the model as needed
* Document and communicate
  Clearly explain the formulation.

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SET-B

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① Apply exhaustic search method to maximize $f(x) = \frac{50x}{x^2+4}$ on the interval $[0,10]$ using 10 equal divisions

Given,

$$f(x) = \frac{50x}{x^2+4} \cdot [0,10] = n = 10$$

$$\Delta x = \frac{b-a}{n} = \frac{10-0}{10} = \frac{10}{10} = 1$$

1) $x_1 = a = 0 \Rightarrow f(x_1) = f(0) = 0$

2) $x_2 = x_1 + \Delta x = 0+1 = 1 \Rightarrow f(x_2) = f(1) = 10$

3) $x_3 = x_2 + \Delta x = 1+1 = 2 \Rightarrow f(x_3) = f(2) = 12.5$

$f(x_1) < f(x_2) > f(x_3)$

Terminate

$\therefore$ min $f(x_1, x_3)$

$(0,2)$ //.

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② A factory employee unskilled workers earning Rs. 135 per week and skilled workers earning Rs. 270 per week. It is required to keep the weekly wage bill below Rs. 24,000. The machines require a minimum of 110 operators, of whom at least 40 must be skilled. union regulations require that number of skilled workers should be at least half the number of unskilled workers. If x is the number of unskilled workers and y the number of skilled workers, develop all constraints to be satisfied by x and y.

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let,

x = unskilled workers

y = skilled workers

1) weekly wage bill constraints

$$135x + 270y \leq 24300$$

2) minimum no. of operators

$$x + y \geq 110$$

3) minimum no of skilled workers

$$y \geq 40$$

4) union regulation

$$y \geq \frac{1}{2}x$$

5) non-negative constraints

$$x \geq 0, y \geq 0$$

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③ Identify and classify the critical points of

$$f(x) = 3x^4 - 24x^3 + 66x^2 - 72x.$$

Given,

$$f(x) = 3x^4 - 24x^3 + 66x^2 - 72x$$

$$\nabla f(x) = 12x^3 - 72x^2 + 132x - 72$$

$$= 12(x^3 - 6x^2 + 17x - 6) = 0$$

$$x^3 - 6x^2 + 11x - 6 = 0$$

$$(x-1)(x-2)(x-3) = 0$$

$$x_1 = 1, x_2 = 2, x_3 = 3$$

$$\nabla^2 f(x) = 36x^2 - 144x + 132$$

$$\Rightarrow \nabla^2 f(x)|_{x_1=1} = 36 - 144 + 132 = 24 > 0 \Rightarrow \text{minima}$$

$$f_{min} = f(x=1) = 3 - 24 + 66 - 72 = -27$$

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$\Rightarrow \nabla^2 f(x)|_{x_2=2} = 144 - 288 + 132 = -12 < 0 \Rightarrow$ maxima

$f_{max} = f(x=2) = 3 \times 16 - 24 \times 8 + 66 \times 4 - 72 \times 2$

$= 48 - 192 + 264 - 144$

$= -24$

$\Rightarrow \nabla^2 f(x)|_{x_3=3} = 36 \times 9 - 144 \times 3 + 132$

$= 324 - 432 + 132 = 24 > 0 =$ minima

$f_{min} = f(x=3) = 3 \times 81 - 24 \times 27 + 66 \times 9 - 72 \times 3$

$= 243 - 648 + 594 - 216$

$= -27.$

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(4) A base station uses two transmit antennas with power levels x_1 & x_2. The total power consumption is $f_1(x) = 4x_1 + 3x_2$ and coverage quality is modelled as $f_2(x) = -x_1^2 - 2x_2^2 + 8x_1 + 6x_2$. subject to $x_1 \geq 0, x_2 \geq 0$. Determine the set of pareto-optimal solution using weighted sum approach.

Given,

$f_1(x) = 4x_1 + 3x_2$

$f_2(x) = -x_1^2 - 2x_2^2 + 8x_1 + 6x_2$

constraints :- $x_1 \geq 0, \; x_2 \geq 0$

w.k.T

utility function $U = \alpha f_1 + (1-\alpha)(-f_2)$

$U(x_1, x_2) = \alpha(4x_1 + 3x_2) + (1-\alpha)(x_1^2 + 2x_2^2 - 8x_1 - 6x_2) \rightarrow (1)$

To find stationary points :-

$\frac{\partial U}{\partial x_1} = 0 \Rightarrow \alpha(4) + (1-\alpha)(2x_1 - 8) = 0$

$4\alpha + (1-\alpha)(2x_1 - 8) = 0 \rightarrow (2)$

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$$\frac{\partial U}{\partial x_2} = 0 \Rightarrow \alpha(2) + (1-\alpha)(4x_2-6) = 0$$

$$3\alpha + (1-\alpha)(4x_2-6) = 0 \rightarrow (3)$$

simplify & find x_1 & x_2 interms of 'α' from (2)

$$4\alpha + 2x_1 - 8 - 2\alpha x_1 + 8\alpha = 0$$

$$12\alpha - 2\alpha x_1 + 2x_1 - 8 = 0$$

$$12\alpha - 2\alpha x_1 = 8 - 2x_1$$

$$\alpha(12 - 2x_1) = 8 - 2x_1$$

$$\alpha = \frac{8-2x_1}{12-2x_1} \rightarrow (4)$$

from (3)

$$3\alpha + 4x_2 - 6 - 4\alpha x_2 + 6\alpha = 0$$

$$9\alpha + 4x_2 - 6 - 4\alpha x_2 = 0$$

$$\alpha(9 - 4x_2) = 6 - 4x_2$$

$$\alpha = \frac{6-4x_2}{9-4x_2} \rightarrow (5)$$

from 4 & 5

$$\frac{8-2x_1}{12-2x_1} = \frac{6-4x_2}{9-4x_2}$$

$$(8-2x_1)(9-4x_2) = (6-4x_2)(12-2x_1)$$

$$\cancel{72} - 32x_2 - 18x_1 + \cancel{8x_1x_2} = \cancel{72} - 12x_1 - 48x_2 + \cancel{8x_1x_2}$$

$$-32x_2 - 18x_1 = -12x_1 - 48x_2$$

$$-32x_2 + 48x_2 = -12x_1 + 18x_1$$

$$16x_2 = 6x_1$$

$$8x_2 = 3x_1$$

$$\boxed{x_1 = \frac{8}{3}x_2} \rightarrow (6)$$

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$\alpha \in [0,1]$, $\alpha \geq 0$

$$\frac{6-4x_2}{9-4x_2} \geq 0$$

$$6-4x_2 \geq 0$$

$$4x_2 \leq 6$$

$$x_2 \leq 3/2$$

$$\boxed{\therefore\ 0 \leq x_2 \leq 3/2}$$

from (5)

$$x_2 = 0 \Rightarrow \alpha = \frac{6-0}{9-0} = \frac{6}{9} = \frac{2}{3}$$

$$x_2 = \frac{3}{2} \Rightarrow \alpha = \frac{6-4\times\frac{3}{2}}{9-4\times\frac{3}{2}} = \frac{6-6}{9-6} = \frac{0}{3} = 0$$

$$\alpha \in \left[0, \frac{2}{3}\right]$$

from (6)

$$x_2 = 0 \Rightarrow x_1 = \frac{8}{3}(0) = 0$$

$$x_2 = \frac{3}{2} \Rightarrow x_1 = \frac{8}{3} \times \frac{3}{2} = 4$$

The points are $(0,0)$, $(4, 3/2)$

Sub $x_2 = 0$ in eq (1)

$$U = \alpha(4x_1 + 3x_2) + (1-\alpha)(x_1^2 + 2x_2^2 - 8x_1 - 6x_2)$$

$$U = \alpha 4x_1 + (1-\alpha)(x_1^2 - 3x_1)$$

a.w.r.to 'x'

$$4\alpha + (1-\alpha)(2x_1 - 8) = 0$$

$$4\alpha + 2x_1 - 8 - 2\alpha x_1 + 8\alpha = 0$$

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$4\alpha + 2x_1 - 8 - 2\alpha x_1 + 8\alpha = 0$

$12\alpha - 2\alpha x_1 = 8 - 2x_1$

$-2\alpha x_1 + 2x_1 = 8 - 12\alpha$

$12\alpha - 8 = 2\alpha x_1 - 2x_1$

$12\alpha - 8 = x_1(2\alpha - 2)$

$x_1 = \frac{12\alpha - 8}{2\alpha - 2}$

$x_1 \geq 0$

$\frac{12\alpha - 8}{2\alpha - 2} \geq 0$

$12\alpha - 8 \geq 0$

$12\alpha \geq 8$

$\alpha \geq \frac{2}{3}$

At $\alpha = \frac{2}{3} \Rightarrow x_1 = \frac{12 \times \frac{2}{3} - 8}{2 \times \frac{2}{3} - 2} = \frac{\frac{24}{3} - 8}{\frac{4}{3} - 2} = \frac{\frac{24-24}{2}}{\frac{4-6}{2}} = 0$

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(5) Apply dominance logic to find the Pareto-front among the three smart-phones. Phone x (camera = 8, weight = 4), Phone y (camera = 6, weight = 2), phone z (camera = 8, weight = 2). The goal is to maximize camera quality and minimize weight

		Phone x	Phone y		non-dominated set
1)	camera	8	6	→ x	
	weight	4	2	→ y	
		Phone y	Phone z		phone z dominates phone y
2)	camera	6	8	→ z	phone y is eliminated.
	weight	2	2	→ z	
		Phone x	Phone z		phone z dominates ph x
3)	camera	8	8	→ z	phone x is eliminated
	weight	4	2	→ z	

∴ Pareto-front = {phone z}.

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